Obradovic 2021 ccRCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/8 samples (dominant 522→153 cliff, cut at geo-mean 282) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→4.28 of 6); both low-cycling lobes (main body 8,207 / neural-crest terminus 902) carry the axis.

Gate 2 — module geography → GO (non-branching)

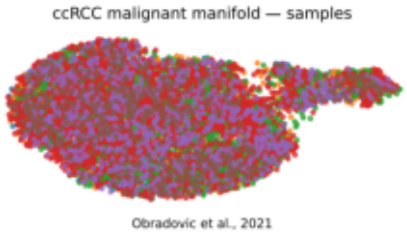
Single dominant pSom→Crest axis: pSom high over the main body declines along pseudotime (ρ=−0.56); Crest anti-correlated (r=−0.95), rising to a terminal readout (ρ=+0.67; Crest-high decile mean pt=0.93). Gast co-varies weakly with pSom (r=+0.47), also declining (early programme, spent); joint pSom&Crest-high depleted (enrichment 0.21) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

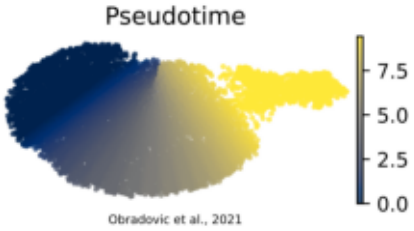
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

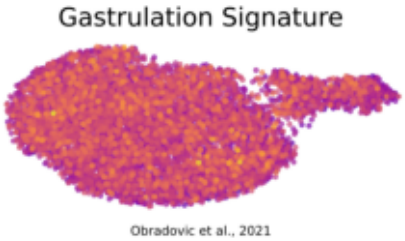
Manifold — samples



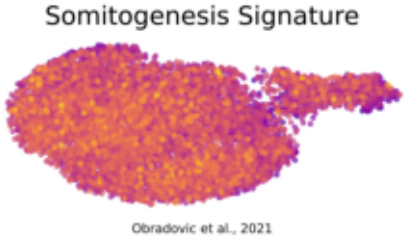
Pseudotime



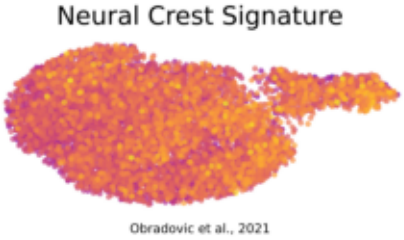
Gastrulation



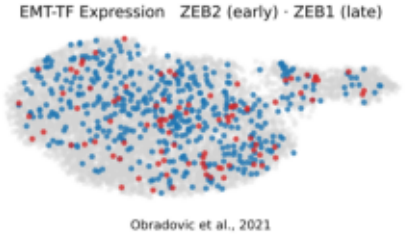
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Clear-cell RCC (Obradovic et al., 2021), 10x — 8 patients / 19,781 cells (master scope); malignant only: 9,369 cells; 9,109 across 6 tumour-bearing samples (drop 2 spillover samples below the 522→153 cliff) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Obradovic's ccRCC malignant manifold is a single monotone somitogenesis→terminal-neural-crest gradient (somitogenesis-high body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Malignant cells are almost entirely non-cycling (~1%), so unlike Kürten/Lin/Qian there is no proliferative-lobe confound: both lobes are low-cycling with near-identical developmental means, and the whole manifold is kept (Bischoff/Sun/Lee/Steele no-scoping precedent). The clean-count-gap premise is met here (dominant cliff 522→153, ratio 3.41 ≫ runner-up 1.86), so the threshold is cut at the geometric mean (282), not the ≥100 floor. EMT-TFs are root/body-loaded: ZEB2 (early) → ZEB1 (late) chosen by data over Puram's SNAI1/ZEB1 pair, but all expressed factors decline along pt and end-separation is weak (Δpt=0.08); no factor rises to the neural-crest terminus.

Parameter choices

Min cells/sample	282 (geo-mean of 522→153 cliff; 6/8)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = pSom-high, Crest-low	**Lobes**	body 8,207 / terminus 902
EMT-TF axis	ZEB2 (early) → ZEB1 (late)	**Geometry**	Linear / non-branching (fold-free)